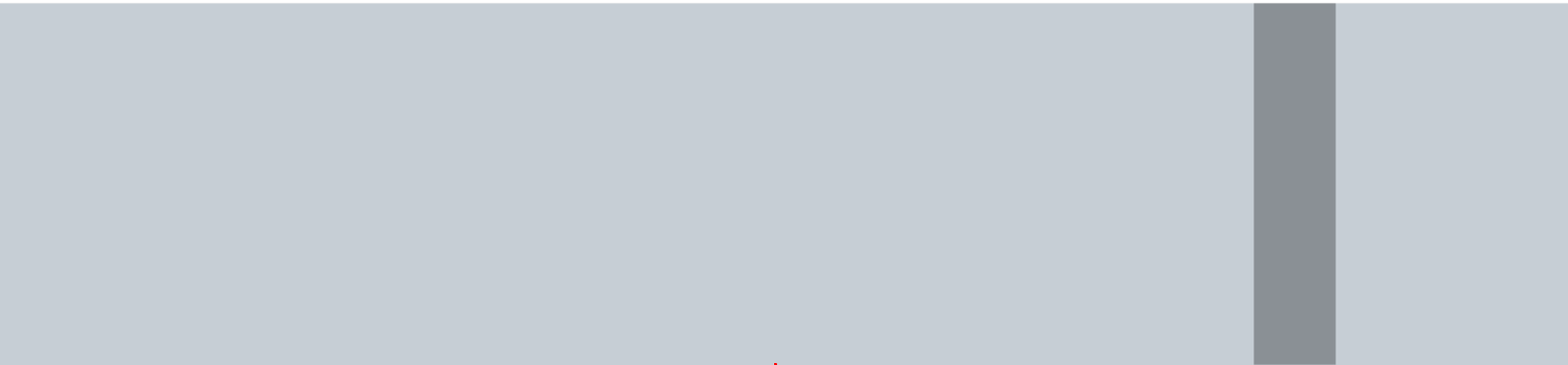
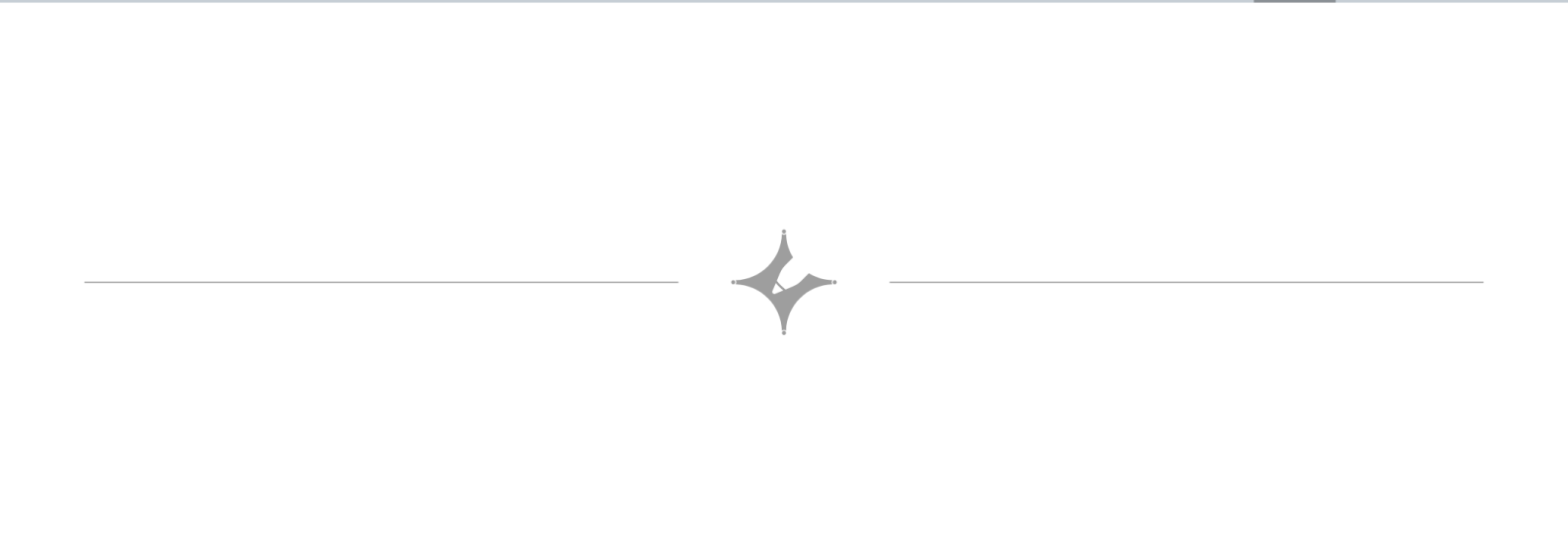
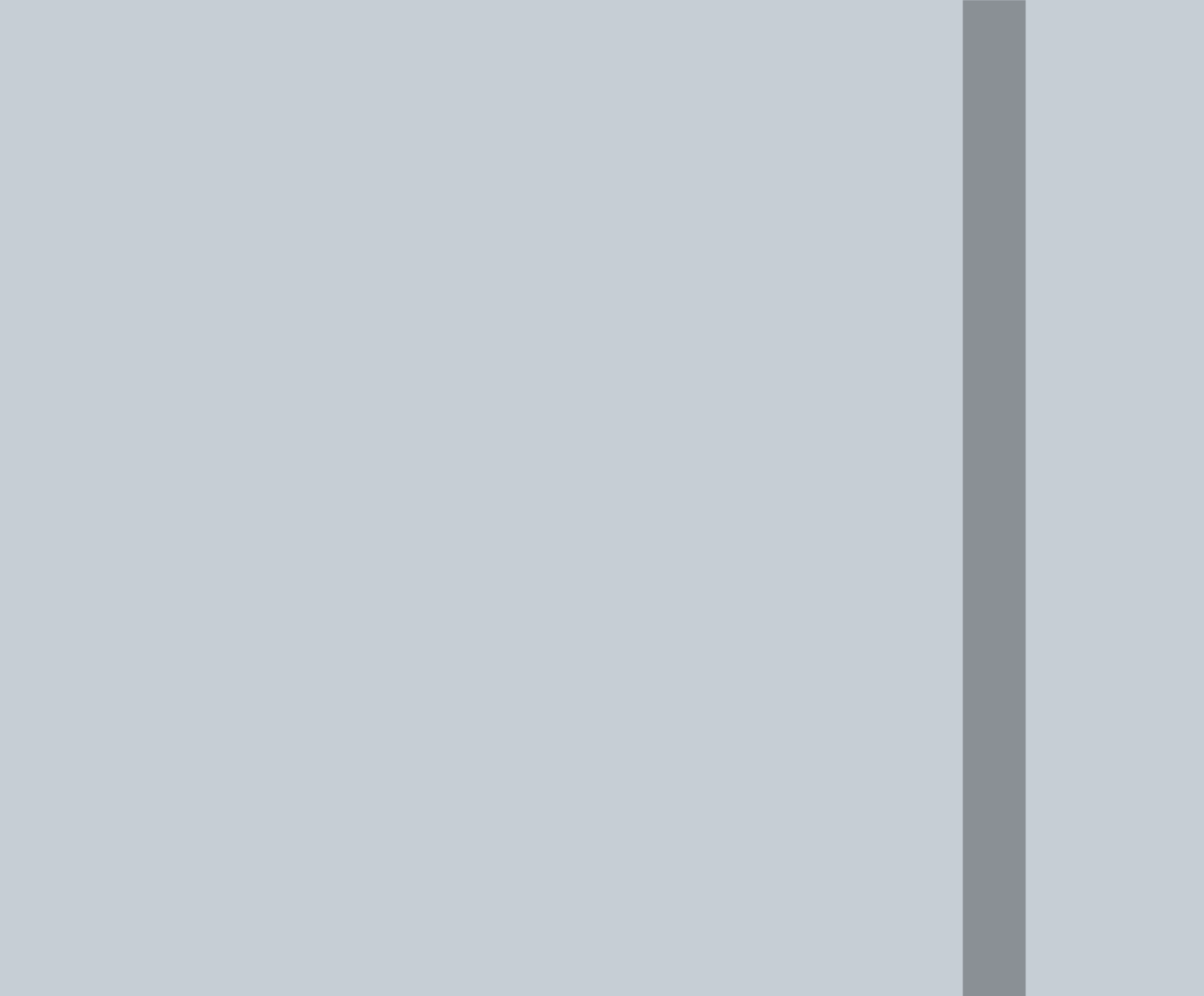




Ratio & proportions

Concept

$$\begin{array}{c} 100 \\ E:A = 2:3 \end{array}$$

$$E = 2x \quad A = 3x$$

$$2x + 3x = 100$$

$$5x = 100$$

$$x = 20$$

$$40 \quad 60$$

$$E = \frac{2}{5} \times 100 = 40$$

$$A = \frac{3}{5} \times 100 = 60$$

Problems of Ratios (ez questions) (LVL-1)

Q5. Rachel purchased one dozen bangles. One day she slipped on the floor and fell down. What cannot be the ratio of broken to unbroken bangles?

~~A) 1:2~~

$$\begin{array}{c} 12 \\ \swarrow \searrow \\ 1:2 \\ 4 \quad 8 \end{array}$$

~~B) 1:3~~

$$\begin{array}{c} 12 \\ \swarrow \searrow \\ 1:3 \\ 3 \quad 9 \end{array}$$

☒ C) 2:3

$$\begin{array}{c} 12 \\ \swarrow \searrow \\ 2:3 \\ 2 \times 12 \\ 5 \end{array}$$

~~D) 1:5~~

$$\begin{array}{c} 12 \\ \swarrow \searrow \\ 1:5 \\ 2 \quad 10 \end{array}$$

Q6. In a mixture of 60 litres, the ratio of milk and water is 2:1. If this ratio is to be 1:2, then the quantity of water to be further added is:

A) 20L

B) 30L

C) 40L

☒ D) 60L

$$\begin{array}{c} 60 \\ M \quad W \\ 2:1 \\ 40 \quad 20 \\ \quad +x \end{array}$$

$$\begin{aligned} \frac{M}{W} &= \frac{40}{20+x} = \frac{1}{2} \\ 80 &= 20+x \\ x &= 60 \end{aligned}$$

(LVL-2) (medium)

Q1. The ratio of work efficiencies of Aquaman and Batman is 3:5 and the ratio of efficiencies of Batman and Cyborg is 5:8. Who is the most efficient?

☒ A) Aquaman B) Batman C) Cyborg D) Can't be determined

$$A:B = 3:5 \times 5$$

$$B:C = 5:8 \times 3$$

$$A:B = 25:15$$

$$B:C = 15:24$$

$$A:B:C = 25:15:24$$

→ but $A:B = 1:2$
 $C:D = 3:4$
 $A:B:C:D =$

Q2. Rs.171 is divided amongst four friends in the ratio $\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{6}$. What is the amount of the greatest share?

A) 14

B) 40

C) 36

☒ D) 60

$$\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{6}$$

$$LCM(3,4,5,6) = 60$$

$$60 \times \frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{6}$$

$$20 : 15 : 12 : 10$$

$$\frac{20}{20+15+12+10} \times 171$$

$$= \frac{20}{57} \times 171$$

$$= 60$$

→ again ratio 3:4:5:6
 total to greatest chunk
 $= \frac{6}{18} \times 171$

but in $\frac{x}{y}$ format $\frac{1}{3}$ will
 be greatest so, either
 $do = \frac{1}{3}$
 $\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} \times 171$

or
 like this both
 are similar.

Q3. Two numbers are in the ratio 3:5. If 9 is subtracted from each, the new numbers are in the ratio 12:23. The smaller number is:

- A) 27 ☒ B) 33 C) 49 D) 55

A) $\frac{3x-9}{5x-9} = \frac{12}{23}$
 B) $\frac{3x-9}{5x-9} = \frac{12}{23}$

$$69x - 9 \times 23 = 60x - 9 \times 12$$

$$69x - 60x = 9 \times 23 - 9 \times 12$$

$$9x = 9(23-12)$$

$$x = 11$$

$$3x = 33$$

→ shortcut

↳ Smaller number is 3 which means that the ans is a multiple of 3, which eliminates 49 & 55. Now, after -ing 9 the new ratio is 12 means the

new number must be a multiple of 12. So, $27 - 9 = 18$ (not a multiple of 12). So, $33 - 9 = 24$ ✓ which is a multiple of 12. So, 12 is the answer.

↓ Same logic

Q5. Two men have income in the ratio 5:3, and their expenditure is in the ratio 8:5. If their savings are in the ratio $\frac{4}{3}$, find the ratio of their combined incomes to their combined expenditures.

- A) 31:32 ☒ B) 32:39 C) 32:33 D) 33:34

	A	B	
Inc	$5x$	$3x = 8x$	$\frac{S_A}{S_B} = \frac{5x-8y}{3x-5y} = \frac{4}{3}$
Exp	$8y$	$5y = 13y$	$15x - 24y = 12x - 20y$
Sav	$5x-8y$	$3x-5y$	$3x = 4y$
			$\frac{x}{y} = \frac{4}{3}$
			$\frac{8 \times \frac{4}{3}}{13 \times \frac{4}{3}} = \frac{32}{39}$

→ Trick

1st ratio should be multiple of 8 (i.e. → 32) and 2nd should be mul. of 13 (i.e. → 39)

Problems of Proportions

(1) $A \propto B$ ✱

means A depends on B.
 $B \uparrow \rightarrow A \uparrow$ but B doesn't depend on A
 If $A \uparrow$ that doesn't mean that B will also increase..

$$\frac{A_1}{A_2} = \frac{B_1}{B_2}$$

$A \propto \frac{1}{B}$ ✱

same with this if $B \uparrow$ then $A \downarrow$ but vice versa is not necessarily true.

$$\frac{A_1}{A_2} = \frac{B_2}{B_1}$$

Q1. A varies directly as B and inversely as C. A is 12 when B is 6 and C is 2. What is the value of A when B is 12 and C is 3?

- ✓ A) 16 B) 12 C) 36 D) 18

$$A \propto \frac{B}{C}$$

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} \times \frac{C_2}{C_1}$$

$$\frac{12}{x} = \frac{6}{12} \times \frac{3}{2}$$

$$x = \frac{12 \times 2 \times 2}{3} = 16$$

Q3. The fourth proportional to 18, 15 and 12 is:

- A) 9 ✓ B) 10 C) 15 D) 21

$$A : B :: C : D$$

$$18 : 15 :: 12 : x$$

$$\frac{18}{15} = \frac{12}{x}$$

$$x = \frac{12 \times 15}{18} = 10$$

Q5. Wayne started a business by investing Rs.36000. After 4 months Stark joined him with some investment. At the end of the year, the total profit was divided between them in the ratio of 9:7. How much capital was invested by Stark in the business?

- A) 21000 B) 42000 C) 38000 D) 84000

$$\text{PROFIT} \propto \text{INVESTMENT} \times \text{TIME}$$

$$\frac{P_w}{P_s} = \frac{I_w}{I_s} \times \frac{T_w}{T_s}$$

$$\frac{9}{7} = \frac{36000}{x} \times \frac{12}{8}$$

$$x = \frac{36000 \times 12 \times 7}{9 \times 8} = 42000$$